

Cart Abandonment Analysis

Business Intelligence Report - Portfolio Project

Checkout funnel performance, abandonment patterns, campaign efficiency, and practical opportunities to improve completed orders and revenue recovery.

Business Question

Where are customers dropping off in the checkout funnel, which abandoned carts are worth recovering, and what should the product and marketing teams fix to recover more revenue?

Executive Summary

- **Total sessions:** 472,871
- **Total orders:** 32,313
- **Conversion rate:** 6.83%
- **Cart abandonment:** 65.97%

The dashboard shows meaningful leakage before cart and during checkout. The largest measured stage-to-stage drop-off is Product to Cart (63.65%), while Cart to Shipping drop-off is 32.09%. Mobile conversion is lower than desktop, and ad conversion rates vary materially. These patterns suggest opportunities to improve product-page engagement, checkout usability, and campaign quality.

Important interpretation: the dashboard's 3.76M "Potential Revenue Lost" is an estimate, not confirmed recoverable revenue. Treat it as a prioritization signal and measure actual recovered orders and revenue through tests.

Scope and Approach

This report summarizes the two Power BI pages supplied for the project: funnel and customer behavior, plus ad/source and product performance. Findings are based on visible dashboard values; underlying event-level data and metric definitions were not independently audited.

1. Checkout Funnel: Where Customers Drop Off

The funnel chart shows the percentage lost between consecutive stages. This is different from a single overall abandonment rate because each percentage uses the previous stage as its denominator.

- **Home to Product - 44.76%:** Many home-stage visitors do not reach a product view.
- **Product to Cart - 63.65%:** Largest stage-to-stage leakage; investigate product-page and add-to-cart barriers.
- **Cart to Shipping - 32.09%:** A sizeable share leaves before the shipping details step.
- **Shipping to Billing - 19.27%:** Some customers leave during checkout progression.
- **Billing to Order - 37.93%:** A notable share does not complete the order after the billing stage.

What this means

Prioritize the Product to Cart transition first because it has the largest observed stage drop-off. Then review Billing to Order and Cart to Shipping. Validate each event definition and ensure stages represent comparable sessions or users before estimating the number of people lost.

Recommended product and checkout actions

- **Product pages:** test clearer pricing, delivery and returns information, product availability, stronger calls to action, and faster page loading.
- **Cart and shipping:** show shipping charges and delivery estimates early; reduce unexpected costs and unnecessary form steps.
- **Billing to order:** review payment failures, payment options, error messages, and checkout friction.
- **Measurement:** track each funnel step consistently and compare abandonment by device, campaign, and new or repeat session.

2. Device and Customer Segment Findings

- **Desktop conversion rate - 8.50%:** Use as a benchmark when investigating the mobile journey.
- **Mobile conversion rate - 3.09%:** Audit mobile product pages, page speed, cart usability, and payment flow.
- **Potential revenue lost, unique sessions - 3.04M (80.89%):** Largest estimated opportunity by segment; test recovery messaging for eligible unique sessions.
- **Potential revenue lost, repeat sessions - 0.72M (19.11%):** Test reminder timing and friction-reducing messages for returning visitors.

Interpretation caution: the customer-segment chart is titled "Potential Revenue Lost by Customers." It supports the conclusion that unique sessions account for more estimated potential loss; it does not, by itself, prove that unique customers generate more completed sales than repeat customers.

3. Marketing and Product Performance

The ad data combines UTM content (individual ad label), source, sessions, orders, and sales. Conversion rates help compare ads, while sessions and sales add scale and context.

- **g_ad_1 (gsearch):** 366,034 sessions, 24,940 orders, sales 1,495,847.24, conversion rate 6.81%
- **b_ad_1 (bsearch):** 54,909 sessions, 3,818 orders, sales 225,563.48, conversion rate 6.95%
- **g_ad_2 (gsearch):** 33,329 sessions, 2,511 orders, sales 151,730.68, conversion rate 7.53%
- **b_ad_2 (bsearch):** 7,914 sessions, 701 orders, sales 43,109.02, conversion rate 8.86%
- **social_ad_2 (socialbook):** 5,590 sessions, 288 orders, sales 18,516.10, conversion rate 5.15%
- **social_ad_1 (socialbook):** 5,095 sessions, 55 orders, sales 3,743.23, conversion rate 1.08%

Marketing insights

- **High-volume driver:** g_ad_1 contributes the most sessions and orders, so even small conversion improvements could matter. Validate campaign spend and profitability before increasing budget.
- **Highest conversion rate:** b_ad_2 is at 8.86%, but it has much lower volume than g_ad_1. Run a controlled test and check cost per order and ROAS before scaling.
- **Social ad gap:** social_ad_1 converts at 1.08%, compared with 5.15% for social_ad_2. Review audience, creative, placement, and landing page; pause or revise only after checking spend and statistical reliability.
- **Brand vs nonbrand:** compare conversion rate, revenue per session, cost per order, and ROAS by campaign type. Session volume alone does not establish which campaign is more valuable.

4. Prioritized Action Plan

- **Priority 1 (High):** Investigate Product to Cart drop-off (63.65%) and test product-page improvements. *Success measure:* product-to-cart rate, add-to-cart events, completed orders.
- **Priority 2 (High):** Audit mobile checkout given 3.09% mobile vs 8.50% desktop conversion. *Success measure:* mobile conversion rate, checkout error rate.
- **Priority 3 (High):** Validate the estimated 3.76M potential revenue loss and run a cart-recovery test. *Success measure:* incremental recovered orders and revenue versus a control group.
- **Priority 4 (Medium):** Test b_ad_2 and improve social_ad_1 after checking spend and sample size. *Success measure:* ROAS, cost per order, conversion rate, sales.
- **Priority 5 (Medium):** Review Billing to Order and Cart to Shipping friction. *Success measure:* stage conversion rates, payment failure and checkout exit rates.

5. Data Quality Checks and Limitations

The dashboard indicates that orders stop appearing after May 2015. This is an important observation, but it should be validated before treating it as a real business outcome.

- **Check source data coverage:** confirm whether sessions, orders, and sales records exist after May 2015 and whether the dataset ends or changes at that point.
- **Check tracking:** verify order events, date relationships, filters, and whether the order measure responds correctly to the date slicer.
- **Check metric definitions:** confirm whether sessions, cart views, shipping views, and orders are session-based, event-based, or distinct-user counts.
- **Check revenue-loss estimate:** document the calculation and avoid presenting the 3.76M estimate as guaranteed revenue that can be recovered.
- **Check campaign profitability:** ad spend is not shown in the supplied dashboard, so conversion rate and sales cannot establish ROAS or profit.

6. Conclusion

The strongest immediate opportunity is to reduce leakage from Product View to Cart, followed by improving mobile conversion and addressing checkout exits. Marketing should protect high-volume traffic while testing higher-converting ads and improving the weakest social ad. Cart-recovery initiatives should be evaluated with incremental recovered orders and revenue, not potential revenue alone.

Project takeaway: Use funnel data to identify friction, segment the opportunity, run controlled changes, and measure real incremental impact.

Dashboard metrics summarized from the supplied Power BI screenshots. Values and metric definitions should be reconciled with the underlying model before publication.